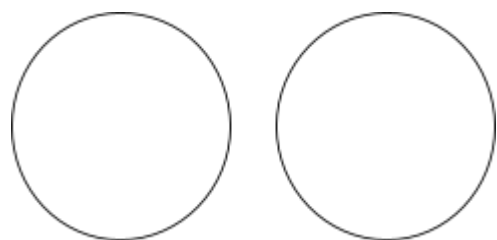


Lesson Goal: By the end of this lesson I will understand what a limit is and be able to calculate the limit of a function given its graph.

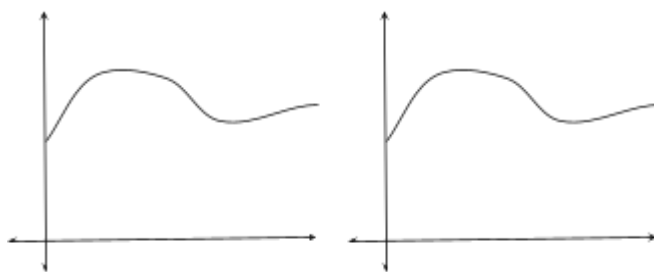
Calculus is the study of _____.

An extremely brief history of ancient calculus

Ancient Greeks used the idea of “limits” to determine the area of a circle by cutting it up into polygons.



Ancient Egyptians had to figure out areas of land around the Nile river to determine taxation. They split up the messy shape around the Nile into rectangles.



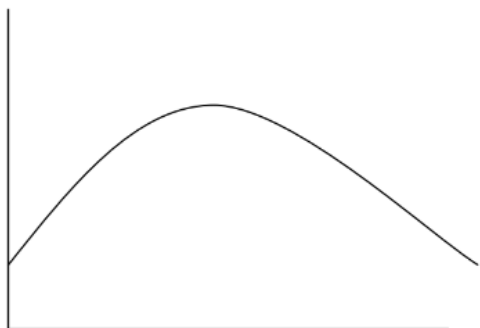
Calculus originated from trying to answer 2 main questions.

a) The Area Problem

b) The Tangent Problem

The **tangent** to a curve at a point P is the straight line through P that most resembles the curve at P. Our goal is to find the **rate of change (slope of the tangent)** at different given points.

But to determine the slope you need _____ points.



To get the second point we are searching for a point Q close to point P.

The **slope of a tangent** to a curve at a point P is the limiting slope of the secant PQ as the point Q slides along the curve towards P. In other words, the **slope of the tangent** is said to be the **limit** of the **slope of the secant** as Q approaches P along the curve.

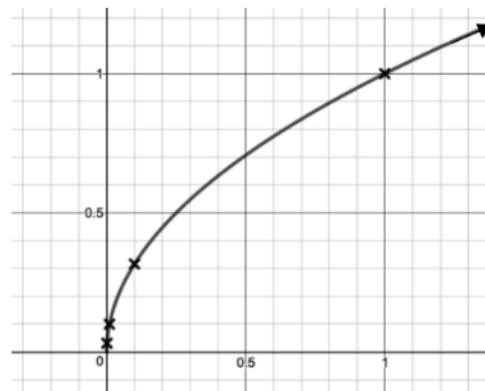
Limits are basic to all of calculus and the process of finding the limits of functions is very important.

The limit of a function is the y-value (behaviour) that a function takes **near** a particular x-value.

The statement, $\lim_{x \rightarrow a} f(x) = l$, is read as: "The limit of the function $f(x)$ as x approaches a is l ."

Example: Determine the limit of the function $y = \sqrt{x}$ as x approaches 1.

The LIMIT of a function is not concerned with the exact y value of a function at a given x value, but rather the TENDENCY in the y values as x approaches a specified value.



Left and Right Hand Limits

These limits are used when the behaviour of the function is different on either side of the indicated x-value.

a) Left hand limit: approach $x = 1$ from the left:

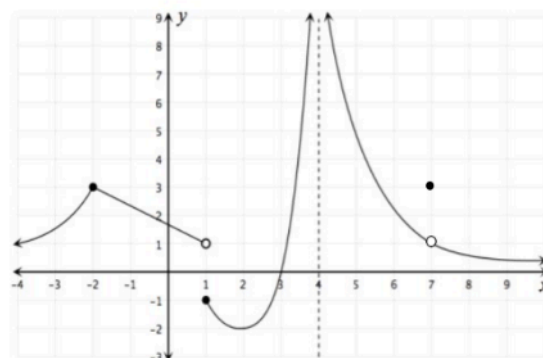
$$\lim_{x \rightarrow 1^-} f(x) =$$

Right hand limit: approach $x = 1$ from the right:

$$\lim_{x \rightarrow 1^+} f(x) =$$

b) $\lim_{x \rightarrow 4^-} f(x) =$

$$\lim_{x \rightarrow 4^+} f(x) =$$



When does a limit exist?

$\lim_{x \rightarrow a} f(x)$ exists if and only if the following three criteria are met:

1. $\lim_{x \rightarrow a^-} f(x) =$ exists
2. $\lim_{x \rightarrow a^+} f(x) =$ exists
3. $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$

Returning to the prior example, this means that

$$\lim_{x \rightarrow 1^-} f(x) =$$

$$\lim_{x \rightarrow 1^+} f(x) =$$

Example: Use the function $y = f(x)$ defined by the following graph, to find each limit.

a) $\lim_{x \rightarrow 4^-} f(x) =$

e) $\lim_{x \rightarrow -1^+} f(x) =$

b) $\lim_{x \rightarrow 2^-} f(x) =$

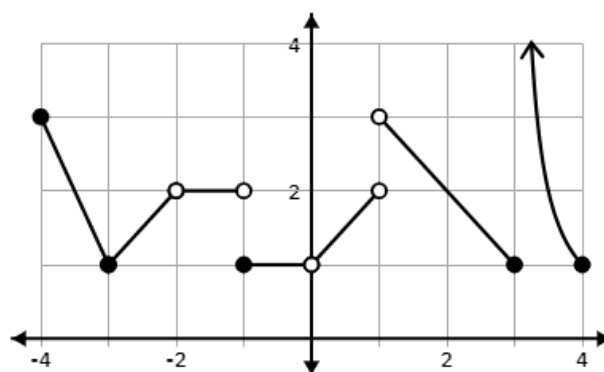
e) $\lim_{x \rightarrow 3^+} f(x) =$

c) $\lim_{x \rightarrow 1^-} f(x) =$

f) $\lim_{x \rightarrow 4^+} f(x) =$

d) $\lim_{x \rightarrow 3^-} f(x) =$

g) $\lim_{x \rightarrow -2^+} f(x) =$

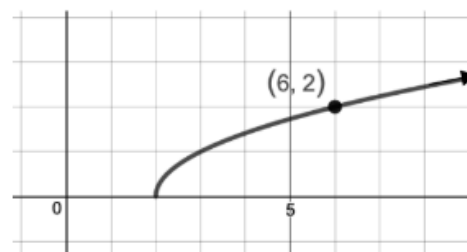


Types of Limits

1. FINITE LIMITS

$\lim_{x \rightarrow a} f(x) = l$ means that $f(x)$ is close to the number l .

This is the most common type of limit.



2. INFINITE LIMITS

a. always occur for a finite x -value (i.e. around vertical asymptotes)

b. $\lim_{x \rightarrow a} f(x) = \pm\infty$ means that $f(x)$ grows without bound,

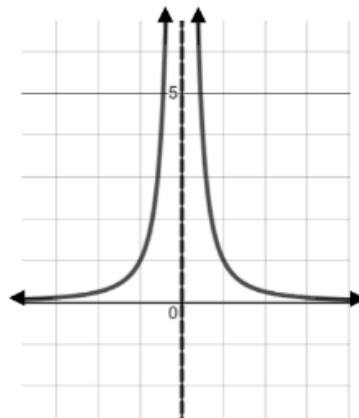
c. eventually becoming bigger/smaller than any number you can name

Remember, ∞ is not a number!

Rather, ∞ is a process of growth that never ends.

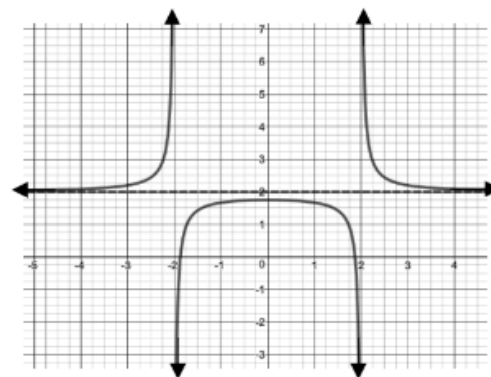
This means that the limit does **NOT** exist. The limit simply expresses the overall trend of the function.

(i.e. $\frac{1}{x^2}$ can be made as large as we like by taking x close enough to 0.)



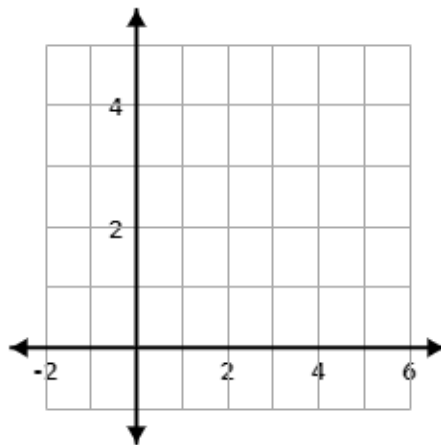
3. LIMITS TO INFINITY: always occur as x -values get very big (very small),

$x \rightarrow \pm\infty$ (i.e. around horizontal asymptotes)



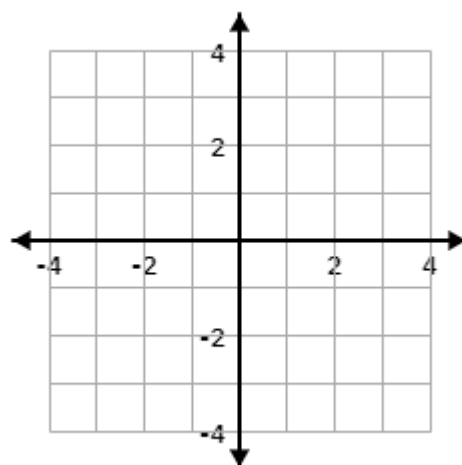
Piecewise Functions

Example 1: Sketch and write the function $f(x) = |x - 2|$ in piecewise form.



Example 2: Sketch and write the function $f(x) = \frac{|x-2|}{x-2}$ in piecewise form.

Then use the graph to determine $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2} f(x)$



Example 3: Sketch and write the function $f(x) = \frac{x^2 - x - 2}{|x - 2|}$ in piecewise form.

Then use the graph to determine $\lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2} f(x)$

